Project Details:

Project Title - CrowdCount-People Counting Using Video Analytics

Project Purpose -

- Detect, track, and count people in real time from live or recorded video feeds.
- Help organizations and event managers monitor crowd density and ensure safety compliance.
- Provide smart alerts, visual dashboards, and live statistics for better crowd management and decision-making.

Tools and Technologies - Python, Flask, OpenCV, YOLOv8, SQLite, JavaScript, Chart.js

Mentor - Ms.Umme Asma S

Batch Number – 3

Start Date - 22-September-2025

Intern Name - U Surya Sasikanth

GitHub Repository -

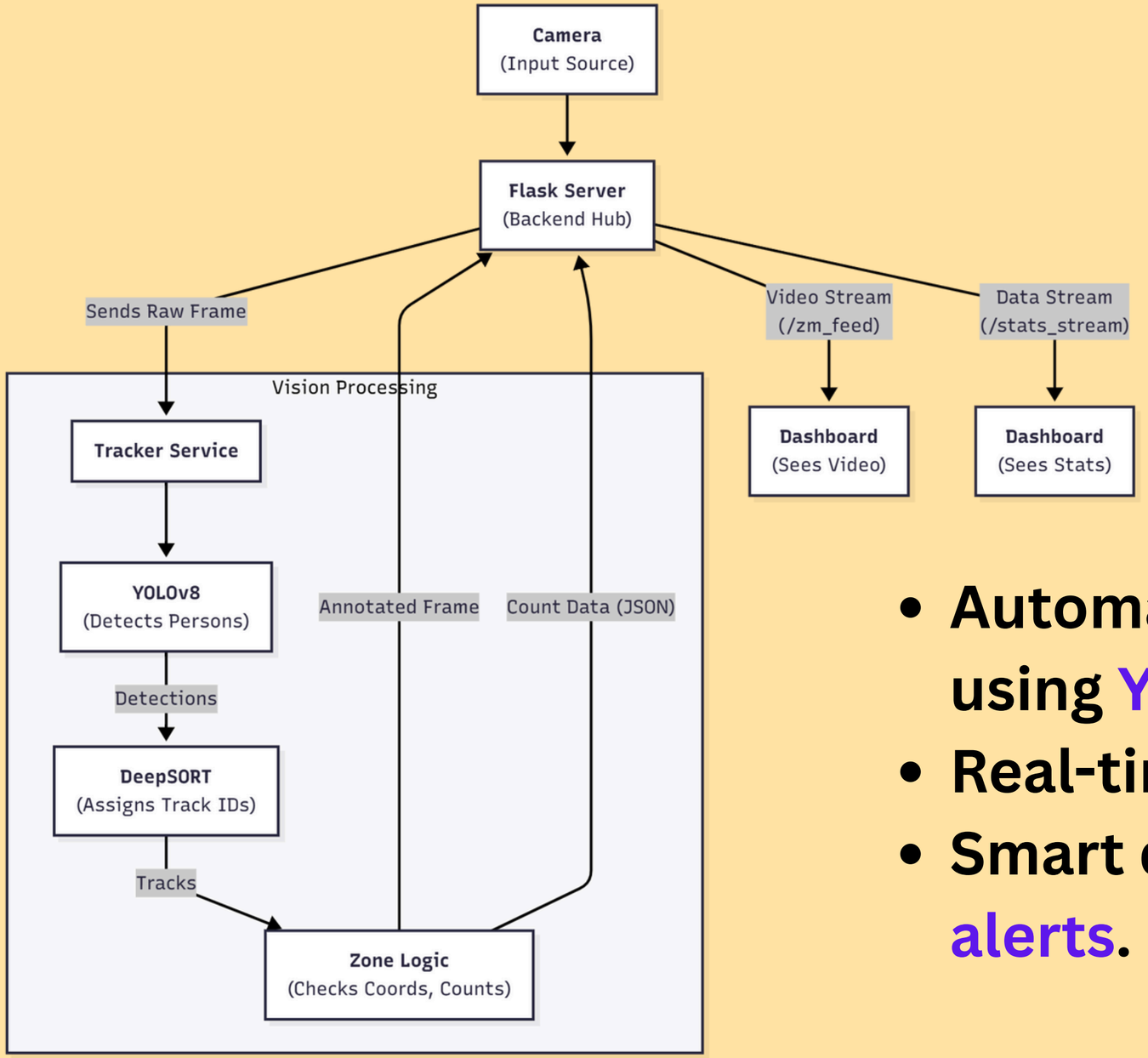
https://github.com/sasi-upparapalli/infosys_crowdcount_october2025_sasi-upparapalli

Project Overview-

https://sasi-upparapalli.github.io/Crowd-Count/

Problem Statement

- ⚠️ Manual crowd monitoring is inefficient and labor-intensive.
- 🕒 Lack of real-time data leads to delayed safety actions.
- 👁️ No accurate way to track occupancy dynamically in public or closed spaces.
- 💰 High cost and human error in traditional crowd management systems.
- 🔐 Limited integration with smart surveillance networks reduces scalability.



Proposed Solution

- Automated detection and counting using YOLOv8.
- Real-time tracking with Deep SORT.
- Smart dashboard for live stats and alerts.

Real-World Implementation Areas

Area of Implementation	Use Case / Purpose
🎓 Educational Institutions	Monitor crowd density in campuses or exam halls.
🏬 Shopping Malls & Public Spaces	Manage foot traffic and ensure safety compliance.
🏢 Corporate Buildings	Track occupancy in lobbies, corridors, and meeting areas.
🚉 Transport Hubs	Monitor passenger flow in stations, airports, and terminals.
🏟️ Events & Stadiums	Maintain safe crowd limits during large gatherings.
🚓 Smart City Surveillance	Integrate with CCTV for real-time crowd analytics.

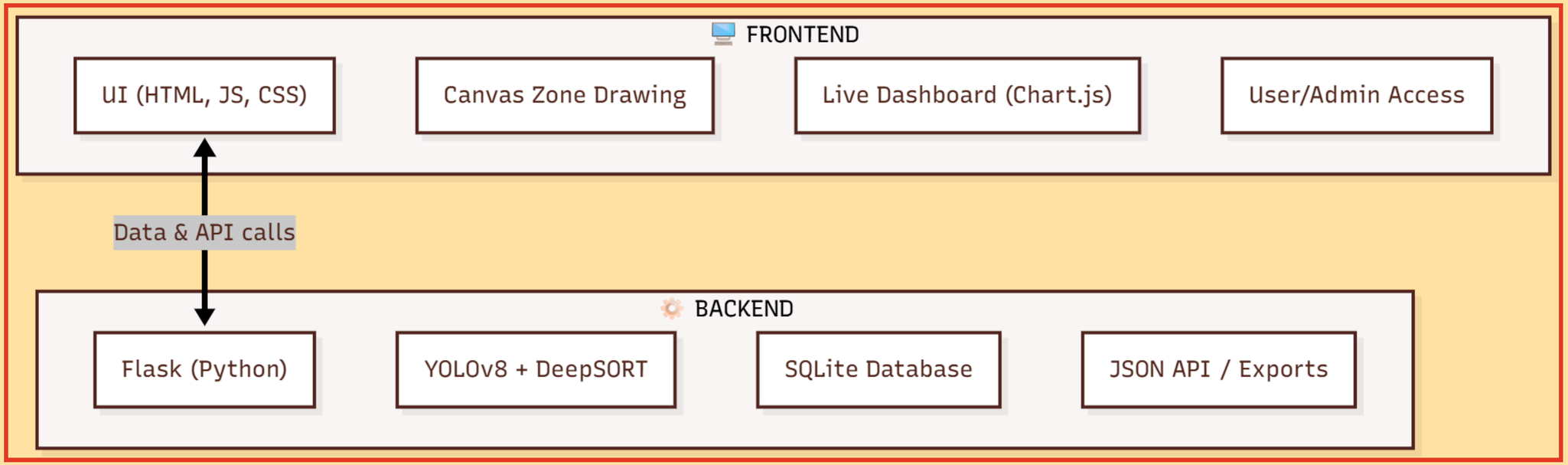
Project Architecture

Frontend

- Built using HTML, CSS, and JavaScript for an interactive user interface.
- Canvas API used for drawing and managing detection zones.
- Chart.js visualizes real-time crowd data and alerts.
- Provides user dashboards and admin panels for control and monitoring.

Backend

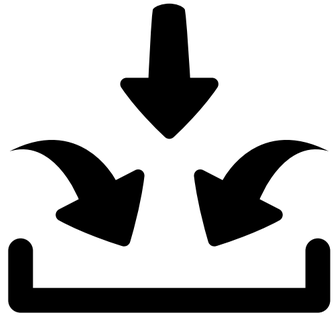
- Developed using Flask (Python) as the main web server.
- Handles video stream processing and model inference (YOLOv8 + Deep SORT).
- Communicates with SQLite3 database to save zones, logs, and user data.
- Provides REST APIs for uploading videos, tracking, exporting reports, etc.



Tech Stack

Category	Tools & Technologies Used
Programming Language	Python, JavaScript
Backend Framework	Flask
Frontend Technologies	HTML5, CSS3, Chart.js, Canvas API
Machine Learning / AI Model	YOLOv8 (Object Detection), Deep SORT (Tracking)
Database	SQLite3
Visualization & UI	Chart.js, Flask Templates (Jinja2)
Libraries & Packages	OpenCV, Ultralytics, FPDF, JWT, Werkzeug
Version Control & Deployment	Git, Render / Local Server
Environment	Python Virtual Environment (venv)

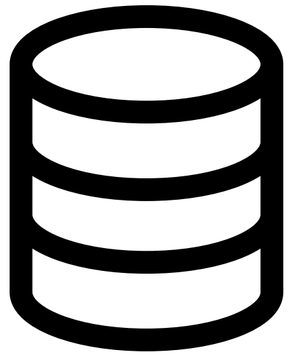
System Workflow



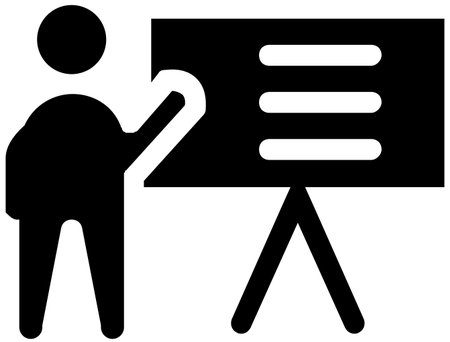
📹 Input Layer – Captures live or uploaded video feeds as the main data source.

⚙️ Processing Layer – Flask routes frames to YOLOv8 for detection; Deep SORT tracks individuals with unique IDs.

🧠 Analytics Layer – Counts people per zone, updates in real time, and triggers alerts on threshold breach.



💾 Storage Layer – Saves users, zones, logs, and reports in an SQLite database.



📊 Presentation Layer – Live dashboard displays analytics, charts, and admin controls.

Key Features of CrowdCount

Role-Based Access Control – Secure login with admin and user privileges

Live People Counting – Real-time detection and tracking using YOLOv8.

Threshold Alerts – Notifies users when a zone exceeds its defined capacity.

Custom Zone Creation – Draw and manage multiple regions of interest on video feeds.

Interactive Dashboard – Visualizes data using Chart.js (line & bubble charts).

Real-Time Updates – Automatic refresh of zone counts on the dashboard within seconds.

Data Export – Download analytics and logs in CSV or PDF format.

Multi-Camera Integration – Add or manage multiple IP or webcam feeds.

Execution Steps

Setup Environment

- Install dependencies using pip install -r requirements.txt and ensure Python 3.9+ with webcam or video input.

Initialize Database

- Run python app.py to auto-create video_zone.db and configure admin credentials.

Load / Stream Video

- Use webcam or upload video from dashboard to start live feed.

Detection & Tracking

- YOLOv8 detects people; Deep SORT tracks unique IDs in real time.

Live Counting & Alerts

- Zones update counts automatically and trigger alerts on threshold breaches.

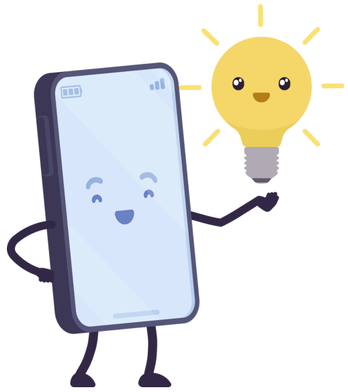
View Dashboard & Export

- Monitor analytics, logs, and export data (CSV/PDF) from the dashboard.

Mathematics & Formulas Behind the Project

Concept / Component	Mathematical Formula / Logic	Purpose in Project
Object Detection (YOLOv8)	**P(Class) × IoU — Combines class probability and Intersection over Union to detect people accurately.**(tracker_service.py)	**Identifies people in each video frame using bounding boxes.**
IoU (Intersection over Union)	**IoU = (Area of Overlap) / (Area of Union)** (tracker_service.py)	**Measures how well the detected bounding box matches the actual object.**
Centroid Tracking	**$C(x, y) = ((x_1 + x_2)/2 , (y_1 + y_2)/2)$** (tracker_service.py)	**Finds the center point of each detected person to track movement.**
Distance Measurement	**$D = \sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2)}$**(tracker_service.py)	**Calculates the movement distance between frames for unique ID tracking.**
People Counting Logic	**Count = Σ (Unique IDs in Zone)** (tracker_service.py)+(script.js)	**Counts unique individuals entering or present in a defined zone.**
Zone Density	**Density = (People Count) / (Zone Area)**(app.py)	**Helps estimate crowd density for safety and alert generation.**
Alert Threshold	**If Count ≥ Threshold → Trigger Alert** (app.py)+(Live.html)	**Sends warning when people count exceeds predefined safe limit.**

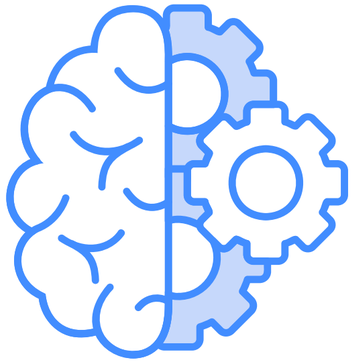
Future Enhancements



Develop a mobile friendly dashboard accessible via smartphones for real-time monitoring and alerts on the go.

Add multilingual options, voice assistance, and accessibility features to make the system inclusive and usable by diverse users.

Add automated voice or sound alerts when crowd thresholds are exceeded, improving real-time responsiveness.



Implement deep learning models to automatically identify unusual crowd movements, panic situations, or suspicious activities in real time.


Thank you